

**MATHEMATICS 1**

**40 MINUTES**

**INSTRUCTIONS TO CANDIDATES**

Please read these instructions carefully, but **do not open this question paper until you are ready to begin.**

This paper contains **27 multiple-choice questions**. There are no penalties for incorrect responses, only marks for correct answers, so you should attempt all of the questions. Each question is worth one mark.

For each question, choose the one option you consider correct. If you make a mistake, erase thoroughly and try again.

You must complete the paper within the time limit.

You can use the question paper for rough working. Dictionaries and calculators are **NOT permitted**.

**Please wait until you are ready before turning this page.**

*Original ESAT CAMP practice material. Not an official UAT-UK or Pearson paper.*

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## **Mathematics 1**

- 1 A graphic designer creates a square logo of side length  $L$ .  
The logo is first reduced in size so that its side length decreases by 25%.  
Realising the logo is too small, the designer attempts to restore it to its original size by increasing the side length of the reduced logo by 25%.  
The area of this final square logo is  $31 \text{ cm}^2$  smaller than the area of the original square logo.  
What is the perimeter of the original square logo?
- A 32 cm  
B 48 cm  
C 60 cm  
D 64 cm  
E 256 cm
- 2 A set of numbers is  $S = \{1, 2, 3, 4, 5\}$ . A number is chosen at random from the set  $S$  three times, with replacement.  
What is the probability that the product of the three chosen numbers is an even number?
- A  $\frac{8}{125}$   
B  $\frac{27}{125}$   
C  $\frac{54}{125}$   
D  $\frac{98}{125}$   
E  $\frac{117}{125}$   
F  $\frac{1}{2}$
- 3 The value of an asset increases by 60%. The new value then decreases by 25%. The final value of the asset is 1080.  
What was the initial value of the asset?
- A 540  
B 800  
C 864  
D 900  
E 1296

- 4 A car travels a certain distance at 30 km/h. It then travels twice that distance at 90 km/h. What is the average speed of the car for the whole journey?

**A** 40 km/h  
**B** 45 km/h  
**C** 54 km/h  
**D** 60 km/h  
**E** 70 km/h

- 5 A gear train consists of four gears: A, B, C, and D.

Gear A meshes with gear B.

Gears B and C are fixed to the same axle.

Gear C meshes with gear D.

Gears B and C are uniform circular discs made of the same material and have the same constant thickness.

The radii of gears A and D are  $r_A = 10$  cm and  $r_D = 20$  cm, respectively.

The masses of gears B and C are  $m_B = 1$  kg and  $m_C = 64$  kg, respectively.

Gear A rotates at 160 revolutions per minute (rpm).

What is the rotational speed of gear D in rpm?

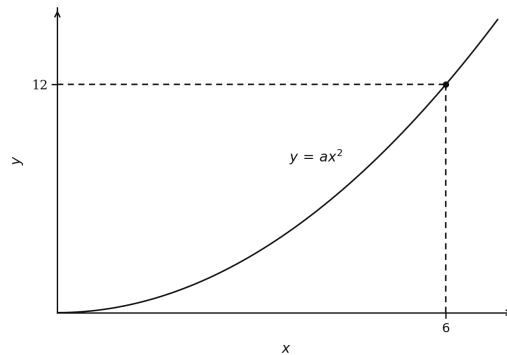
**A** 10  
**B** 40  
**C** 320  
**D** 640  
**E** 2560  
**F** 5120

- 6 The diagram shows the graph of  $y = ax^2$  for  $x \geq 0$ , where  $a$  is a constant. The curve passes through the point  $(6, 12)$ .

The variable  $x$  is inversely proportional to  $t$ .

When  $t = 4$ ,  $x = 3$ .

Which of the following gives an expression for  $y$  in terms of  $t$ ?



- A  $y = \frac{4}{t}$
- B  $y = \frac{4}{t^2}$
- C  $y = \frac{3}{16t^2}$
- D  $y = \frac{48}{t^2}$
- E  $y = \frac{288}{t^2}$
- F  $y = \frac{432}{t^2}$
- G  $y = \frac{3t^2}{16}$
- H  $y = 48t^2$
- 7 A composite cylindrical rod has length  $L$  and total radius  $2r$ . It consists of a solid inner core of radius  $r$  and a surrounding outer sleeve. The core is made of a material with density  $3\rho$  and the sleeve is made of a material with density  $\rho$ . Which expression gives the total mass of the rod?

- A  $4\pi r^2 L \rho$
- B  $6\pi r^2 L \rho$
- C  $7\pi r^2 L \rho$
- D  $8\pi r^2 L \rho$
- E  $10\pi r^2 L \rho$
- F  $12\pi r^2 L \rho$

- 8 A digital file of size 120 Megabytes is to be transmitted over a network. The transmission protocol requires that for every 15 parts of data, 1 part of overhead is added.
- The network has a constant transmission speed of 32 Megabits per second.
- Assuming 1 Megabyte =  $10^6$  bytes, and 1 byte = 8 bits, what is the total time required to transmit the file and its overhead?
- A 0.5 seconds  
 B 2 seconds  
 C 4 seconds  
 D 30 seconds  
 E 32 seconds  
 F 480 seconds
- 9 Pipe P can fill a tank in 10 hours. Pipe Q can fill the same tank in  $T$  hours.
- Pipe P works alone for 2 hours to begin filling the empty tank. Then pipe Q is also opened, and working together, they take a further 4 hours to fill the tank.
- What is the value of  $T$ ?
- A 5  
 B 7.5  
 C 8  
 D 10  
 E 15  
 F 20
- 10 Two quadratic equations of the form  $x^2 + 2bx + c = 0$  are formed.
- For each equation, the coefficients  $b$  and  $c$  are chosen independently and uniformly at random from the set  $\{1, 2, 3\}$ .
- What is the probability that at least one of the two equations has real roots?
- A  $\frac{4}{81}$   
 B  $\frac{28}{81}$   
 C  $\frac{49}{81}$   
 D  $\frac{7}{9}$   
 E  $\frac{8}{9}$   
 F  $\frac{77}{81}$

- 11 The finite region  $R$  is bounded by the four lines with equations:

$$3x + 4y = 22$$

$$3x - 4y = 14$$

$$3x + 4y = -2$$

$$3x - 4y = -10$$

What is the area of  $R$ ?

- A 12
  - B 20
  - C 24
  - D 25
  - E 48
- 12 The positive integers  $x, y, z$  have no common factor strictly greater than 1. They satisfy the relationship:

$$3x : 4y : 5z = 2 : 3 : 4$$

Given that  $k$  is a constant such that  $kx + y - z = 77$ , what is the value of  $k$ ?

- A 2
  - B 26
  - C 27
  - D 38
  - E 39
  - F 40
- 13 Seven distinct items are arranged in a row. How many arrangements are possible in which no two of three specific items are adjacent?
- A 240
  - B 576
  - C 720
  - D 1440
  - E 4320

- 14** A curve is given by the equation  $y = (x - k)(x^2 - 4)$ , where  $k$  is a real constant such that  $k > 1$ .  
 A dataset consists of the  $y$  -intercept and all the  $x$  -intercepts of the curve.  
 The mean of this dataset is equal to its median.

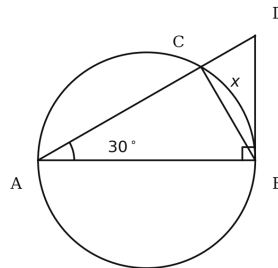
What is the value of  $k$ ?

- A**  $\frac{2}{9}$
- B**  $\frac{4}{7}$
- C**  $\frac{4}{3}$
- D**  $\frac{8}{5}$
- E**  $\frac{8}{3}$
- F** 6

- 15 The diagram shows a circle with diameter  $AB$ .

The straight line  $AD$  passes through point  $C$  on the circumference of the circle, and the line  $BD$  is tangent to the circle at  $B$ .

Given that  $\angle CAB = 30^\circ$  and the length of  $BC$  is  $x$ , which of the following is a correct expression for the length of  $CD$  in terms of  $x$ ?



[diagram not to scale]

- A  $\frac{1}{3}x$
- B  $\frac{1}{2}x$
- C  $\frac{\sqrt{3}}{6}x$
- D  $\frac{\sqrt{3}}{3}x$
- E  $\frac{\sqrt{3}}{2}x$
- F  $\frac{2\sqrt{3}}{3}x$
- G  $\sqrt{3}x$
- H  $2\sqrt{3}x$

- 16** Two cones have the same volume. The height of the second cone is twice the height of the first cone.

If the base radius of the first cone is  $r$ , what is the base radius of the second cone?

- A**  $\frac{r}{\sqrt{2}}$
- B**  $\frac{r}{2}$
- C**  $\frac{r}{4}$
- D**  $r\sqrt{2}$
- E**  $2r$
- F**  $4r$

- 17** The curve  $C$  has equation  $y = 6x - x^2$ .

The point  $M$  is the maximum point of  $C$ .

The points  $P$  and  $Q$  have position vectors  $\vec{OP}$  and  $\vec{OQ}$  such that:

$$3\vec{OM} = \vec{OP} + 2\vec{OQ}$$

where  $O$  is the origin.

Given that  $P$  has coordinates  $(7, 3)$ , what is the distance from  $Q$  to the point where  $C$  intersects the positive  $x$ -axis?

- A** 13
- B**  $\sqrt{13}$
- C**  $\sqrt{73}$
- D**  $\sqrt{145}$
- E**  $\sqrt{229}$
- F**  $4\sqrt{37}$

- 18** Liquid X has a density of 4.  
Liquid A has a density of 5.  
Liquid B has a density of 2.

A mixture, Y, is created by combining liquids A and B. A new blend is then formed by mixing liquid X and mixture Y.

In this final blend, the ratio of the volume of X to the volume of Y is 1 : 3.

The ratio of the mass of X to the mass of Y is 2 : 5.

What is the ratio of the volume of liquid A to the volume of liquid B in mixture Y?

- A** 1 : 1
- B** 2 : 1
- C** 4 : 5
- D** 5 : 4
- E** 14 : 1
- F** 4 : 25

- 19** A fair six-sided die is rolled.  
If the number rolled is a multiple of 3, a fair coin is tossed 3 times.  
Otherwise, the coin is tossed 2 times.  
What is the probability of obtaining exactly 2 heads?

- A**  $\frac{5}{24}$
- B**  $\frac{13}{48}$
- C**  $\frac{7}{24}$
- D**  $\frac{5}{16}$
- E**  $\frac{1}{3}$
- F**  $\frac{3}{8}$

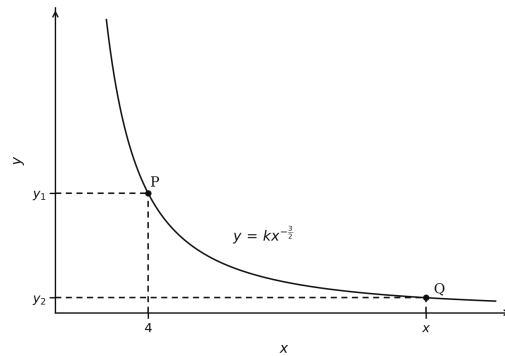
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The diagram shows part of the curve with equation  $y = kx^{-\frac{3}{2}}$ , where  $k$  is a positive constant and  $x > 0$ .

The points  $P(4, y_1)$  and  $Q(x, y_2)$  lie on the curve.

The value of  $y_2$  is 87.5% less than the value of  $y_1$ .

What is the value of  $x$ ?



- A 8
- B  $8\sqrt{2}$
- C 16
- D 32
- E 64

21

Three fair, six-sided dice are rolled. Let  $p$  be the probability that the highest value shown is 4. Let  $q$  be the probability that the lowest value shown is 2.

What is the value of  $q - p$ ?

- A 0
- B  $\frac{7}{72}$
- C  $\frac{1}{9}$
- D  $\frac{37}{216}$
- E  $\frac{61}{216}$

- 22** In a mathematical investigation, a shape is defined by the equation:

$$x^2 + y^2 - 4tx - 2ky + 4t^2 = 0$$

The parameter  $k$  is a fixed positive constant. The parameter  $t$  is a variable, and  $t > 0$ .

Which of the following statements about the shape is/are correct as  $t$  varies?

1. The area of the shape is constant.
2. The shape has exactly one point of intersection with the  $x$ -axis.
3. The distance of the center of the shape from the origin is a linear function of  $t$ .

- A** 1 only  
**B** 2 only  
**C** 1 and 2 only  
**D** 1 and 3 only  
**E** 2 and 3 only  
**F** 1, 2 and 3

- 23** A composite solid sphere of outer radius  $2r$  consists of a central spherical core of radius  $r$  and density  $\rho_c$ , surrounded by a shell of density  $\rho_s$ .

The mean density of the composite sphere is  $2\rho_s$ .

What is the ratio  $\frac{\rho_c}{\rho_s}$ ?

- A** 5  
**B** 7  
**C** 8  
**D** 9  
**E** 15

- 24** A data set  $S$  contains  $n$  distinct integers, where  $n \geq 2$ . Let  $\mu$  be its mean,  $m$  its median and  $r$  its range. A further integer data value  $k$  is appended to the data set; repeated values are allowed after this addition.

Which statements must be true?

1. If  $k > \mu$ , the new mean is greater than  $\mu$ .
2. If  $k > r$ , the new range is greater than  $r$ .
3. If  $k < m$ , the new median is less than  $m$ .

- A** 1 only  
**B** 1 and 2 only  
**C** 1 and 3 only  
**D** 2 and 3 only  
**E** 1, 2 and 3

- 25** For a regular polygon with  $n$  sides, the angle between a side extended and the adjacent side is  $(k - 20)^\circ$ , where  $k$  is a constant.

Which of the following expressions gives the value of  $n$ ?

- A**  $\frac{180}{k - 20}$   
**B**  $\frac{360}{k - 20}$   
**C**  $\frac{180}{200 - k}$   
**D**  $\frac{360}{200 - k}$

- 26** The variables  $x$  and  $y$  are such that the graph of  $y$  against  $x$  is a straight line through the origin.  
When  $x = 2$ ,  $y = 8$ .

The constants  $p$  and  $q$  are given by  $p = 5 \times 10^7$  and  $q = 2 \times 10^{-3}$ .

A quantity  $M$  is defined as:

$$M = \frac{\left(\frac{y}{q}\right)}{\left(\frac{x}{p}\right)}$$

What is the value of  $M$ ?

- A**  $1 \times 10^5$
- B**  $6.25 \times 10^9$
- C**  $2.5 \times 10^{10}$
- D**  $1 \times 10^{11}$
- E**  $1.6 \times 10^{-10}$
- F**  $6.25 \times 10^3$

- 27** The real number  $x$  satisfies the equation:

$$\frac{x - \sqrt{3}}{x + \sqrt{3}} = 2 - \sqrt{3}$$

What is the value of  $x$ ?

- A** -3
- B** -1
- C** 1
- D** 2
- E** 3